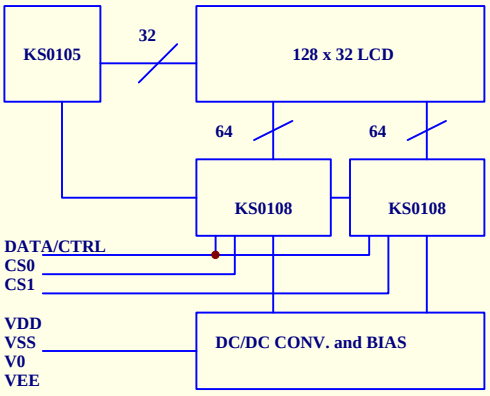
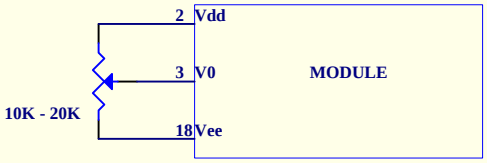
## *128 x 32 Graphic Module*

| Pinout: Pin: | Function | |
|---|---|---|
| 1 | Vss | GND. 0V |
| 2 | Vdd | +5Vdc |
| 3 | V0 | CONTRAST ADJUST INPUT |
| 4 | RS | H= DATA, L= INSTRUCTIONS |
| 5 | R/W | H= READ, L= WRITE |
| 6 | E | ENABLE PULSE |
| 7 | DB0 | DATA BUS |
| 8 | DB1 | |
| 9 | DB2 | |
| 10 | DB3 | |
| 11 | DB4 | |
| 12 | DB5 | |
| 13 | DB6 | |
| 14 | DB7 | DATA BUS |
| 15 | CS1 | RIGHT SIDE CHIP SELECT (SEGS 65 - 128) |
| 16 | CS0 | LEFT SIDE CHIP SELECT (SEGS 1 - 64) |
| 17 | /RESET | L= RESET |
| 18 | Vee | -5Vdc LCD BIAS VOLTAGE |
| 19 | LED+ | Vf =4.2V @ 15mA |
| 20 | LED- | |

### BLOCK DIAGRAM




### CONTRAST ADJUST




| Title | | |
|---|---|---|
| Size A | Number | Revision |
| Date: 12-Mar-2000 | | Sheet of |
| File: C:\PFWSCH\GR12632.SCH | | Drawn By: |

# NOTES ON INITIALIZING THE 128X32 MODULE

**This module uses two KS0108 Ram enabled segment drivers. The KS0108 has four control instructions. They are:**

1. **Display ON/OFF**
2. **Set Address ( Y or column address 0 - 63 )**
3. **Set Page Address ( X address 0 - 7 ) each page is 64 bytes**
4. **Set Start Line ( line at top of screen 0 - 63 )**
   **This instruction can also be used to vertically scroll the display.**
   **There is also a status read with the msb indicating 'BUSY' status.**

**The two KS0108 chips are selected by CS_0 or CS_1.**
**A low on CS_0 selects the left half of the Display (segments 1 - 64).**
**A low on CS_1 selects the right half of the display (segments 65 - 128).**

**The KS0108 Display Data ram stores pixel data for the LCD. It is a 64 Column X 64 Row addressable array, as shown on the attached drawings. The 64 rows are divided into 8 pages of 8 lines. Data is read from or written to the 8 lines of each page directly through DB0 to DB7. On this display only four of the eight pages are**
**Displayed at any time.**

**To Write to the display you must :**
**set a page address**
**set a column address**
**set a start line address**
**begin writing data.**

**The column address increments after a write.**
**A reset does not initialize the display ram.**
**Here is a typical sequence to start the display.**
**(Note: this must be done for both halves of the display)**

| | |
|---|---|
| **3EH** | **Display Off** |
| **0C0H** | **Set Ram Line 0 to be the initial display line** |
| **0B8H** | **Set Page Address (to page 0)** |
| **40H** | **Set (Y) Address = 0** |
| | **Write 64 bytes of data (initial display data or zero's to clear display ram)** |
| | **Repeat for each display page until display full** |
| **3FH** | **Display ON** |

**Use the display start line command to access the rest of the display ram. IE if the start line is 0 then page 0 is at the top of the display. If it is 8 then page 1 is at the top of the display and so on. You can change the start line in increments of one to implement a smooth scroll.**
**Note that the KS0108 is strictly dot matrix graphic and does not have an internal character generator . Any characters displayed must be sent as bitmap graphic information. However this allows you to define any size characters that will fit in the 32 lines.**

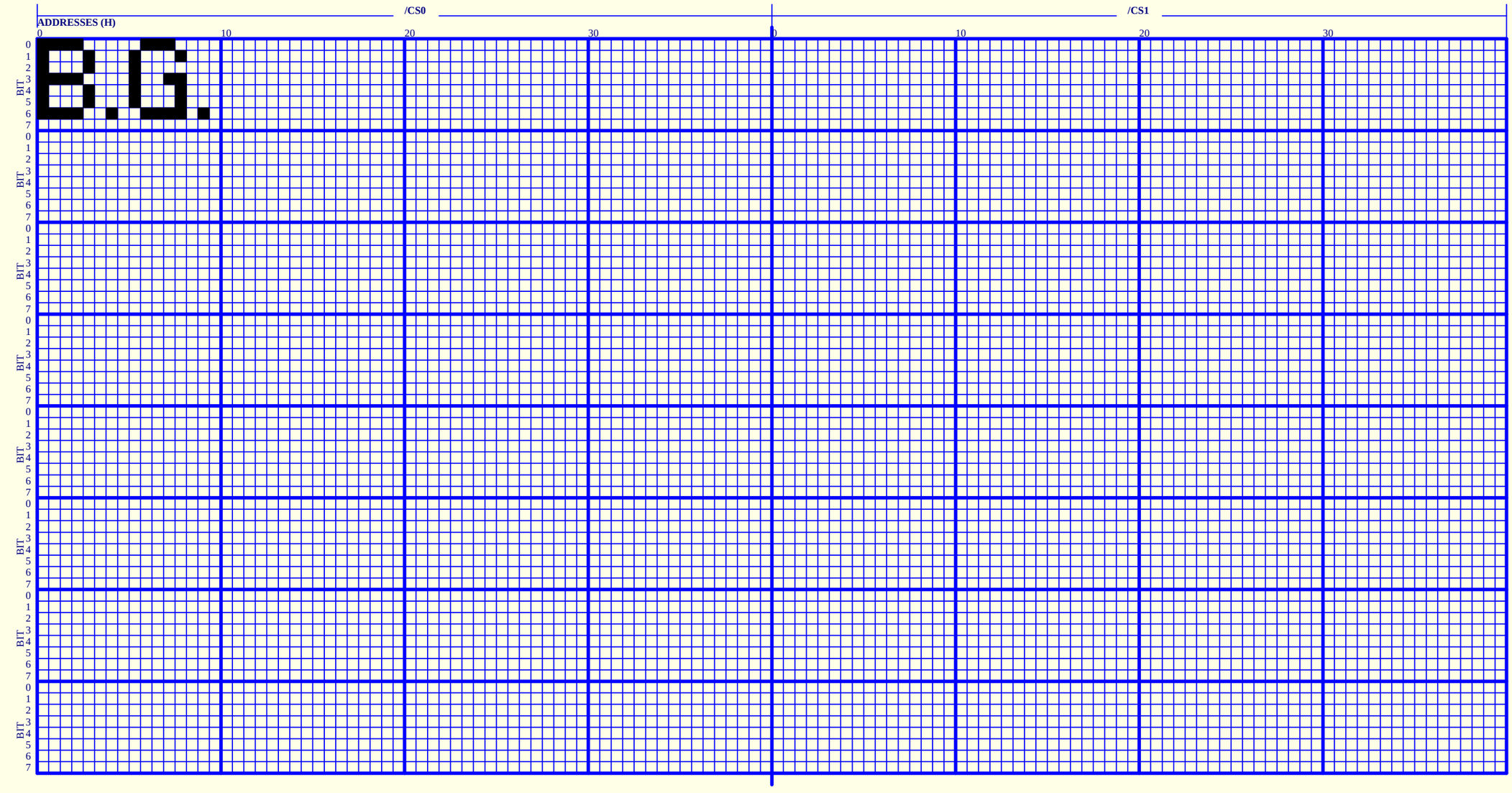
**The LED backlight on this module is Vf = 4.2V @ 15mA.**

| BIT | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 2 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 3 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 4 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 5 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 6 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 0 |
| 7 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

## RAM TO DISPLAY MAPPING

**NOTE:**

**ONLY FOUR RAM PAGES ARE DISPLAYED AT A TIME**




| | | |
|---|---|---|
| Title | | |
| Size: B | Number | Revision |
| Date: 13-Mar-2000 | | Sheet of |
| File: C:\PFWSCH\BG_GRAF.SCH | | Drawn By: |

## DISPLAY CONTROL INSTRUCTION

The display control instructions control the internal state of the KS0108B. Instruction is received from MPU to KS0108B for the display control. The following table shows various instructions.

| Instruction | RS | R/W | DB7 | DB6 | DB5 | DB4 | DB3 | DB2 | DB1 | DB0 | Function |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Display ON/OFF | L | L | L | L | H | H | H | H | H | L/H | Controls the display on or off. Internal status and display RAM data is not affected. L:OFF, H:ON |
| Set Address | L | L | L | H | Y address (0~63) | | | | | | Sets the Y address in the Y address counter. |
| Set Page ( X address) | L | L | H | L | H | H | H | Page (0~7) | | | Sets the X address at the X address register. |
| Display Start Line | L | L | H | H | Display start line (0~63) | | | | | | Indicates the display data RAM displayed at the top of the screen. |
| Status Read | L | H | BUSY | L | ON/OFF | RESET | L | L | L | L | Read status.<br>BUSY L: Ready<br>H: In operation<br>ON/OFF L: Display ON<br>H: Display OFF<br>RESET L: Normal<br>H: Reset |
| Write Display Data | H | L | Write Data | | | | | | | | Writes data (DB0:7) into display data RAM. After writing intruction, Y address is increased by 1 automatically. |
| Read Display Data | H | H | Read Data | | | | | | | | Reads data (DB0:7) from display data RAM to the data bus. |


(C) 1997 SAMSUNG Electronics
Printed in Korea

**(3) MPU Interface**

| Chatacteristic | Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| E Cycle | $t_C$ | 1000 | - | - | ns |
| E High Level Width | $t_{WH}$ | 450 | - | - | ns |
| E Low Level Width | $t_{WL}$ | 450 | - | - | ns |
| E Rise Time | $t_R$ | - | - | 25 | ns |
| E Fall Time | $t_F$ | - | - | 25 | ns |
| Address Set-Up Time | $t_{ASU}$ | 140 | - | - | ns |
| Address Hold Time | $t_{AH}$ | 10 | - | - | ns |
| Data Set-Up Time | $t_{SU}$ | 200 | - | - | ns |
| Data Delay Time | $t_D$ | - | - | 320 | ns |
| Data Hold Time (Write) | $t_{DHW}$ | 10 | - | - | ns |
| Data Hold Time (Read) | $t_{DHR}$ | 20 | - | - | ns |

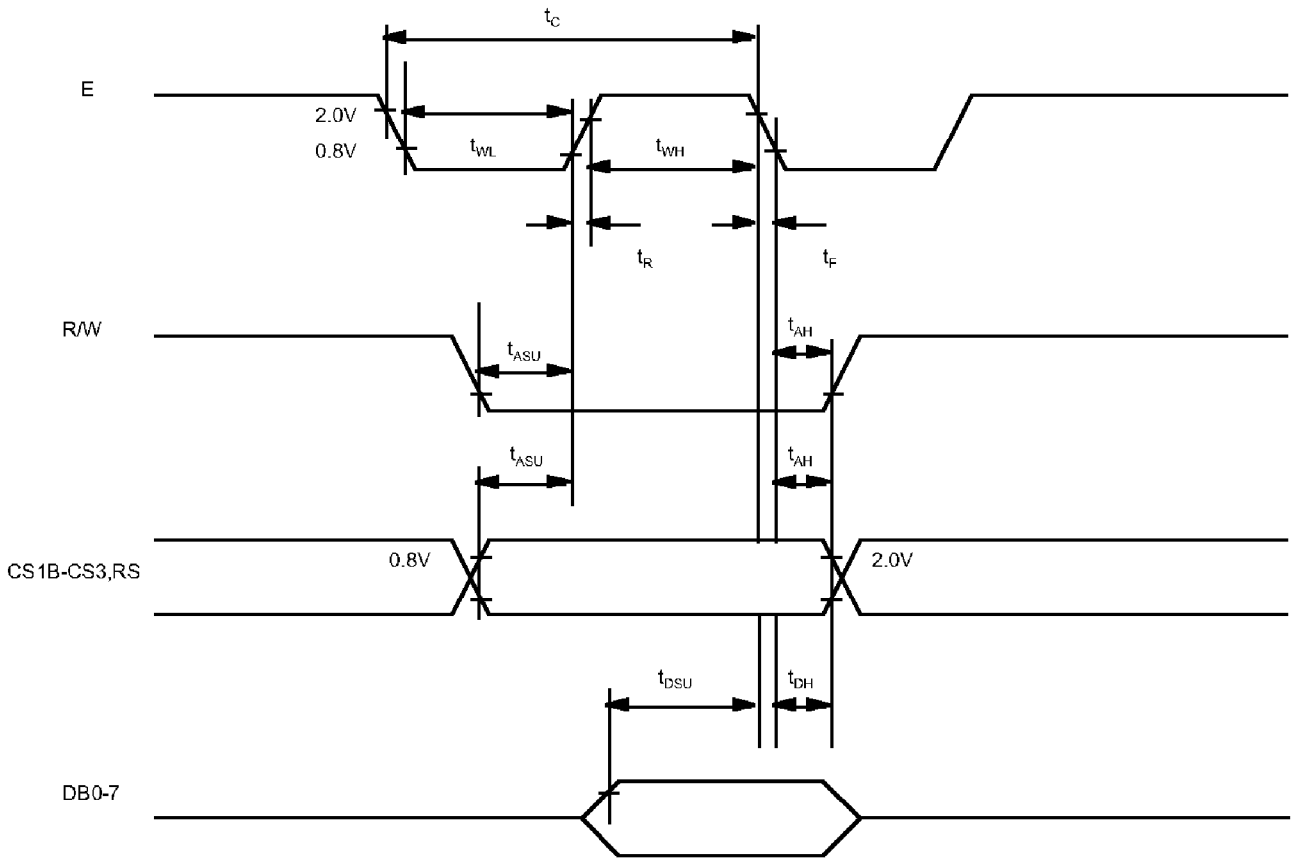



Fig 3. MPU write timing


(C) 1997 SAMSUNG Electronics
Printed in Korea

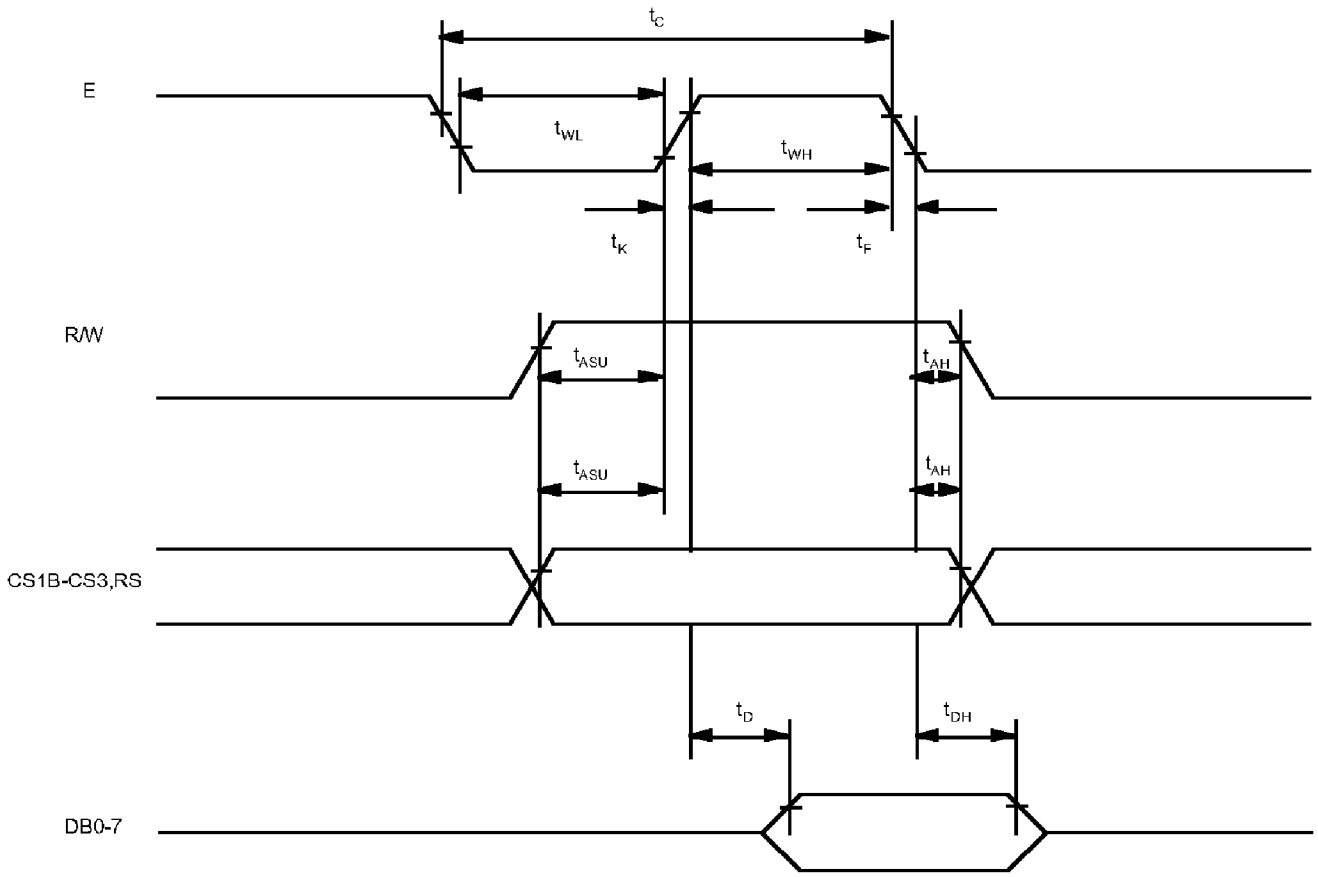



Fig 3. MPU write timing


(C) 1997 SAMSUNG Electronics
Printed in Korea